GFHMP: Gradual Fusion Framework of Hyperspectral, Multispectral, and Panchromatic Images Using a Novel Spatial–Spectral Cross-Fusion Network

Bolin Fu, *Member, IEEE*, Xifeng Deng, Hongyuan Kuang, Zhaoyin Wang, and Donglin Fan

***Abstract*—This article proposes a gradual fusion framework for hyperspectral (LR-HSI, 30 m), multispectral (MR-MSI, 10 m), and panchromatic (HR-PAN, 2.5 m) images (GFHMP) to achieve large-scale multiresolution satellite image fusion. The GFHMP framework employs an innovative spatial–spectral cross-fusion network (SSICFNet) that effectively integrates complementary spatial and spectral information from multiresolution inputs. The proposed SSICFNet utilizes a spectral cross-attention block (SECB) and a spatial cross-attention block (SACB) to fuse LR-HSI with MR-MSI or medium-resolution hyperspectral image (MR-HSI) with HR-PAN images, which effectively integrates spatial–spectral information for high-quality image reconstruction. The performance of GFHMP and SSICFNet was evaluated using both the public simulated Chikusei dataset and a custom real-world M-ZY1E dataset. Experimental results demonstrate that SSICFNet achieves state-of-the-art performance, outperforming existing methods on both datasets. Quantitative evaluation on the Chikusei dataset shows peak signal-to-noise ratio (PSNR) values of 43.94 dB (MR-HSI) and 42.94 dB [high-resolution hyperspectral image (HR-HSI)] for images fused with different spatial resolutions. The reconstructed vegetation spectral curve in HR-HSI exhibits exceptional agreement (R^2 = 0.9993) with the reference data. For the M-ZY1E dataset, the reconstructed images display high spatial consistency with panchromatic image references and strong spectral correlation with original data, confirming the robustness of the GFHMP framework. To compensate for the lack of systematic validation regarding fused image feasibility in quantitative inversion, we employed LR-HSI, MR-HSI, and HR-HSI data from the M-ZY1E dataset to invert canopy chlorophyll content (CCC) in the Qinzhou Coastal Mangrove Reserve, Guangxi, China. The fused image achieves an inversion accuracy of R^2 = 0.759, representing an 11.49%–41.60% improvement over LR-HSI standalone inversion. We employed Shapley additive explanations (SHAP) to quantitatively assess the impacts of enhanced spectral fidelity and spatial resolution on chlorophyll retrieval. SHAP analysis demonstrates that HR-HSI texture features significantly improve the mangrove CCC estimation, validating the advantages of image fusion for quantitative inversion. The source code is publicly available at: https://github.com/RSFubolin/GFHMP**

***Index Terms*—Chlorophyll content inversion, deep learning, explanatory analysis, gradual fusion, satellite hyperspectral image, spatial–spectral information extraction.**

I. Introduction

SATELLITE hyperspectral images have been extensively employed in large-scale remote sensing studies, including image classification [1], [2] and target detection [3], [4], due to their large-area imaging advantages. However, physical limitations inherent to satellite optical sensors [5] restricted most satellite platforms to acquiring hyperspectral imagery with relatively coarse spatial resolution hyperspectral (LR-HSI, 30 m), consequently constraining advanced applications, including inversion vegetation functional traits retrieval and species-level classification. In recent years, a large number of deep learning-based dual-input fusion (DF) methods [6], [7], [8], [9], [10], [11], [12] have been proposed to solve this challenge, which hyperspectral–multispectral fusion or hyperspectral–panchromatic fusion to generate high-resolution hyperspectral image (HR-HSI).

Although the hyperspectral–multispectral fusion method performs excellently in spectral information retention, it has a limited effect on hyperspectral spatial resolution enhancement. While the hyperspectral–panchromatic fusion can provide higher spatial resolution compared with hyperspectral–multispectral fusion, it is prone to obvious distortions in the fused images due to the significant spatial scale difference between hyperspectral and panchromatic images [5]. In order to further maintain spectral information while improving the spatial resolution of satellite hyperspectral images, some scholars have proposed hyperspectral, multispectral, and panchromatic image fusion methods [5], [13], [14]. Yokoya et al. [13] introduced a pioneering coupled nonnegative matrix factorization (CNMF)-based approach for hyperspectral, multispectral, and panchromatic image fusion, while Ma et al. [14] developed an unconstrained least-squares deconvolution framework to address this multisensor fusion

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challenge. HyperNet developed in [5] is the first and currently only deep learning-based hyperspectral, multispectral, and panchromatic image fusion algorithm that can generate HR-HSI with accurate spectral information and clear spatial details on simulated datasets. However, due to the lack of high-resolution training reference images, satellite true images are typically trained using downsampled images as training inputs and the original low-resolution images as references, based on Wald's principle. This training method can achieve excellent results in small-scale fusion, such as the hyperspectral–multispectral fusion. Nevertheless, a greater than tenfold spatial resolution discrepancy exists between the panchromatic and hyperspectral datasets. This huge difference in scale makes it easy for Wald's principal training method to cause spectral distortion in the fusion results, and there is also a problem of insufficient training datasets. To address this issue, we propose a gradual fusion framework of hyperspectral, multispectral, and panchromatic images (GFHMP). This framework leverages a dual-input deep learning model (SSICFNet) to achieve HR-HSI reconstruction via a two-step fusion process.

Existing dual-input fusion models mainly include CNN-based [15], [16] and Transformer-based models [10], [17]. CNN-based models show significant performance improvements over traditional models. However, the limited receptive field and lack of remote modeling capabilities of a convolutional neural network (CNN) make it difficult for CNN-based networks to extract comprehensive features from hyperspectral images. To address this issue, a Transformer-based architecture was proposed. The attention mechanism in the Transformer enables the network to have global interaction capabilities. Hu et al. [18] pioneered the application of Transformer architectures to hyperspectral-multispectral fusion through their development of Fusformer. Ma et al. [19] proposed 3DT-Net, which integrates Transformer and 3D-CNN to learn prior knowledge in hyperspectral images. Although the above Transformer-based methods demonstrate excellent fusion performance, surpassing CNN-based methods, these methods only accept single-modal inputs and are difficult to achieve cross-modal interaction between hyperspectral and multispectral or panchromatic images. Ma et al. [9] proposed a dual cross-attention fusion network called DCTransformer, which uses a multihead cross-attention module to calculate cross attention between hyperspectral and multispectral features as inputs, greatly promoting the fusion of spatial and spectral information between the two images. However, this architecture fails to adequately model the latent interimage relationships between input modalities. To further enhance the dual-input fusion performance, we propose a spatial–spectral cross-fusion network (SSICFNet). This network first extracts joint spatial–spectral features from the input pairs (LR-HSI with MR-MSI or medium-resolution hyperspectral image (MR-HSI) with HR-PAN) and constructs cross-fusion features between them. Then, it employs a triple cross-attention mechanism to enable multilevel feature interaction, which significantly boosts fusion quality. By combining CNN-based feature extraction with Transformer attention mechanisms, SSICFNet outperforms both standalone CNN and Transformer models in fusion accuracy.

The GFHMP proposed in this article performs collaborative fusion of satellite hyperspectral, multispectral, and panchromatic images through a two-step fusion process. Compared with traditional hyperspectral–multispectral fusion and hyperspectral–panchromatic fusion methods, this approach can achieve large-scale reconstruction with higher spectral fidelity. Relative to one-step hyperspectral, multispectral, and panchromatic image fusion models, this framework substantially reduces the required training data volume. To further improve the fusion quality of dual-input models, we also proposed a spatial spectral cross-fusion network (SSICFNet), which incorporates two innovative modules: the spectral cross-attention block (SECB) and the spatial cross-attention block (SACB). These novel modules (SECB and SACB) effectively promote the transmission of spatial, textural, and spectral information between input images. To further evaluate the practical feasibility of GFHMP fused images, this study conducted a joint inversion experiment for estimating mangrove canopy chlorophyll content (CCC) using fusion images (MR-HSI and HR-HSI) and original image (LR-HSI) from the coastal area of Qinzhou City, southeast China. Quantified the contribution of spectral and spatial–textural features in fused imagery to CCC estimation, thereby validating their utility for quantitative remote sensing applications. In addition, by independently inverting CCC using hyperspectral images at different resolutions, we systematically evaluated the influence of spatial resolution on inversion accuracy. The main contributions of this work are as follows.

1) We proposed a gradual fusion framework of GFHMP. Adhering to Wald's protocol, this framework substantially reduces the spatial extent and volume of training data required for large-scale satellite hyperspectral image fusion compared with one-step methods, while mitigating associated spectral distortion. The framework's outputs—fused MR-HSI (10 m) and HR-HSI (2.5 m) images—were rigorously validated using the public simulated Chikusei dataset and the custom real-world M-ZY1E dataset.
2) We designed a novel spatial–spectral cross-fusion network (SSICFNet) to enhance GFHMP output quality. SSICFNet integrates an SECB and SACB to comprehensively extract joint spatial–spectral features. Deep dense residual connections mitigate gradient vanishing during fusion, with trainable weighting coefficients adaptively regulating multilevel feature integration.
3) We innovatively validated GFHMP and SSICFNet performance through the mangrove CCC retrieval. A novel joint inversion framework synergistically leveraging fused and original imagery achieved a 41.60% accuracy improvement over conventional single-source inversion. This study pioneers the application of Shapley additive explanations (SHAP) to quantitatively assess global and local feature contributions in fused imagery, demonstrating its significant potential for remote sensing inversion. This work establishes a foundation for fused imagery in quantitative vegetation parameter retrieval.

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II. Related Works

A. Dual-Input Image Fusion Methods Based on Shallow Learning

Dual-input image fusion originates from multispectral and panchromatic image fusion, also known as pansharpening [14]. The pansharpening fuses the hyperspectral with the panchromatic band extracted from the multispectral to generate an HR-HSI [20]. Subsequently, numerous shallow learning-based dual-input fusion methods have been proposed, including matrix decomposition, tensor decomposition, and Bayesian-based approaches. The matrix factorization method [21], [22], [23] fuses spectral and spatial information by jointly decomposing hyperspectral and multispectral data into endmember and abundance matrices with coupled constraints. In [23], a coupled nonnegative matrix factorization (CNMF) method was proposed to significantly improve the accuracy of pasture vegetation classification by fusing hyperspectral and multispectral data. This approach, while demonstrating improved performance, is computationally intensive and highly sensitive to parameter selection. To mitigate spectral distortion, tensor decomposition-based methods have been proposed [24], [25], [26]. In [25] proposed a rank-adaptive tensor decomposition framework, which automatically learns optimal subspace dimensions from the data. While tensor decomposition excels in capturing multiway correlations, its deterministic nature inherently fails to model spectral uncertainty, resulting in suboptimal fusion for heterogeneous scenes. Bayesian methods, by contrast, mitigate this via probabilistic priors [27]. The first known Bayesian fusion method was proposed by Zhang et al. [28]. This article proposes a noise-resistant wavelet-based Bayesian fusion method to effectively integrate multispectral and hyperspectral images while suppressing noise interference. Despite their promising performance, these methods critically depend on manually designed prior assumptions for the target HR-HSI, often introducing spatial artifacts and spectral distortions in the fused results [29], [30].

B. Dual-Input Image Fusion Methods Based on Deep Learning

Deep learning has demonstrated superior efficiency than conventional model-based approaches for various computer vision tasks, particularly in image classification, object detection, and image fusion [31]. This has motivated extensive research into deep learning methods for fusing hyperspectral with multispectral (or panchromatic) images. Dian et al. [24] combined traditional methods with deep learning. They proposed a hybrid approach that first obtains HR-HSI estimates via Sylvester equation solving, then refines them through residual learning, outperforming traditional methods. To further improve fusion quality, Han et al [32] proposed a multilevel and multiscale spatial–spectral fusion CNN for hyperspectral image super-resolution, which effectively integrates hierarchical features across different levels and scales to enhance the spatial resolution. However, these approaches do not take full advantage of deep end-to-end learning. Qu et al. [33] first introduced an unsupervised approach for hyperspectral–multispectral fusion through their sparse Dirichlet-Net, which automatically extracts spectral features from hyperspectral images and spatial features from multispectral images via a shared decoder, eliminating the need for the labeled training data. Numerous unsupervised methods have emerged [34], [35], [36], which have investigated deep learning applications for hyperspectral–multispectral fusion through specialized network architectures. DBIN [34] proposes a deep learning-based framework for blind hyperspectral image fusion, addressing the challenge of reconstructing HR-HSI without prior knowledge of the degradation process, leveraging advanced neural network architectures to enhance both spatial and spectral fidelities. In [36], a model-guided deep convolutional network (MoG-DCN) is proposed, which combines proximal gradient descent (PGD) with a U-Net-based denoiser by unfolding optimization iterations into a learnable network architecture for hyperspectral image super-resolution. While DCNNs perform well in hyperspectral–multispectral fusion, their structure inherently limits their ability to model long-range dependencies.

C. Hyperspectral, Multispectral, and Panchromatic Images Fusion

Research on the fusion of hyperspectral, multispectral, and panchromatic image remains relatively limited. Previous studies have employed various mathematical frameworks to address this fusion problem. Yokoya et al. [13] employed coupled nonnegative matrix factorization (CNMF) to realize the hyperspectral, multispectral, and panchromatic images fusion. Bendoumi and Benlefki [37] tried solving the fusion problem using an unconstrained least-squares unmixing method. Arablouei [38] formulated the fusion problem as a maximum likelihood estimation task, incorporating observation models and linear mixing assumptions. However, these conventional methods encounter practical limitations: they demand meticulous parameter tuning and rely on assumptions that may not always be valid, ultimately compromising their fusion performance. Li et al. [5] proposed HyperNet, the first end-to-end deep learning framework for simultaneous fusion of hyperspectral, multispectral, and panchromatic images. This unified network architecture achieves superior spectral–spatial consistency compared with conventional fusion methods. However, for satellite hyperspectral images trained using Wald's protocol, the significant spatial resolution difference between hyperspectral and panchromatic images requires substantial downsampling ($\times 12$), leading to frequent spectral distortion in fused results.

III. Methods

A. Gradual Fusion Framework of Hyperspectral, Multispectral, and Panchromatic Images

This article proposes a gradual fusion framework of hyperspectral, multispectral and panchromatic images (GFHMP), which employs a two-step fusion method to reconstruct high-resolution hyperspectral images (HR-HSIs, 2.5m).

1) *Fusion of LR-HSI and MR-MSI [FHM, Fig. 1(a)]:* The adaptive selection deep learning model (SSICFNet) is used to extract MR-MSI spatial–spectral information,

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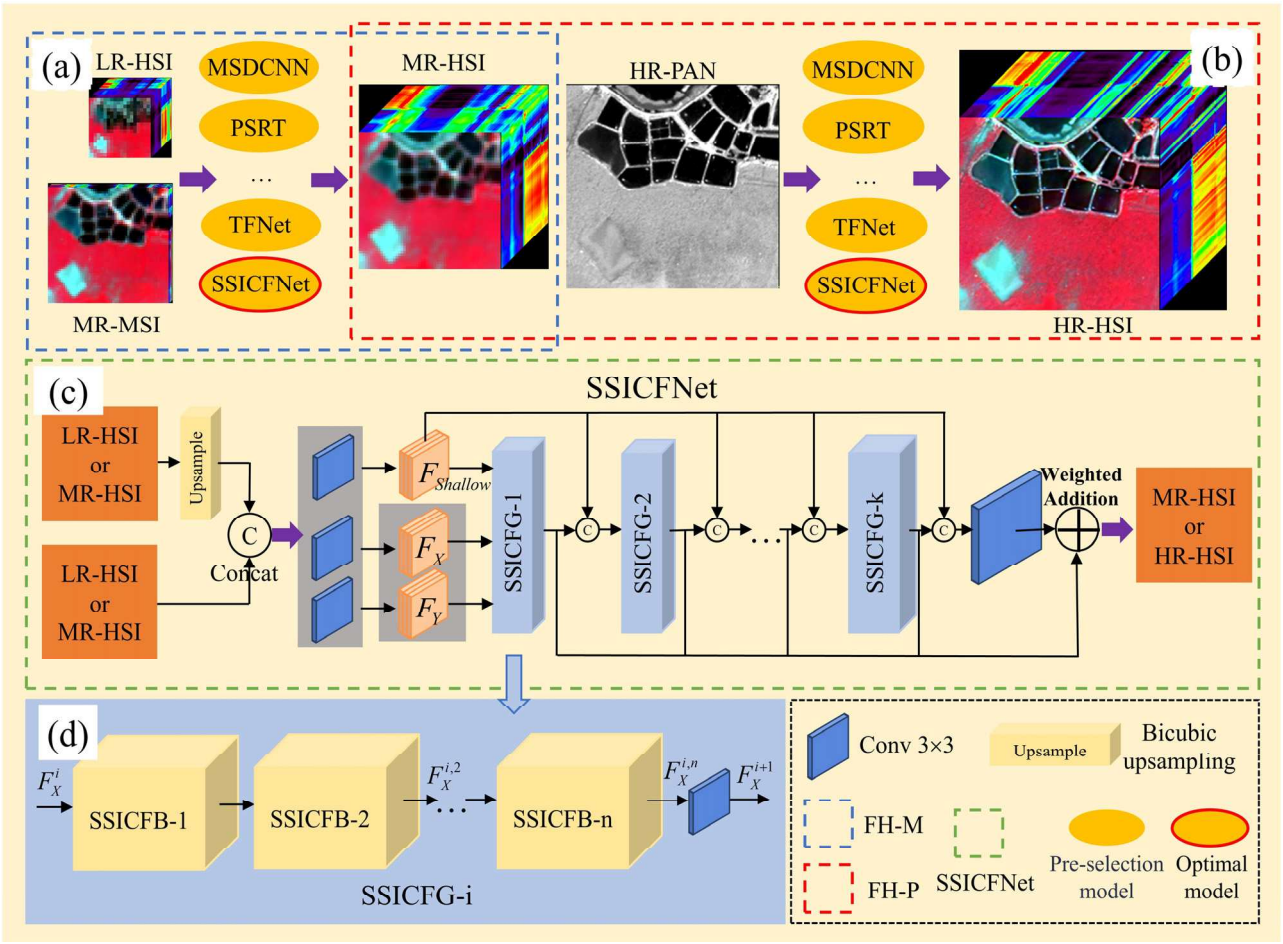


Fig. 1. GFHMP fusion framework and architecture of the proposed SSICFNet. (a) FHM. (b) Fusion of MR-HSI and LR-HSI. (c) Architecture of the proposed SSICFNet. (d) Architecture of the proposed SSICFG-i.

achieving dual optimization of satellite hyperspectral spatial and spectral information to obtain 10-m resolution hyperspectral image (MR-HSI).

2) *Fusion of MR-HSI and HR-PAN [FHP, Fig. 1(b)]:* The adaptive selection deep learning model (SSICFNet) extracts fine spatial information from HR-PAN and injects it into MR-HSI to generate 2.5-m resolution hyperspectral image (HR-HSI). The selected deep learning models are the SSICFNet proposed in this article, two CNN-based models (MSDCNN and TFNet), and three Transformer-based models (MIMFormer, DCTransformer, and PSRT). This framework progressively integrates multisource data, significantly reducing training data requirements while effectively mitigating image distortion, thereby substantially enhancing the fusion quality. The parameter settings for all comparison methods followed the recommendations in the original literature or were optimized through grid search. Both steps of GFHMP fusion were trained according to Wald's principle. It has been verified that SSICFNet has higher fusion performance than other models on FHM and FHP.

Hyperspectral and multispectral images can be regarded as tensors or 3-D data cubes. In this article, they are reshaped into 2-D matrices, and the original satellite hyperspectral, multispectral, and panchromatic images can be represented as $Z \in \mathbb{R}^{wh \times S}$, $Y \in \mathbb{R}^{\mathrm{WH} \times s}$, and $P \in \mathbb{R}^{W_P H_P \times 1}$, respectively. LR-HSI has high spectral resolution but low spatial resolution. S is the number of bands, and w and h are the image width and height, respectively. MR-MSI has both medium spatial resolution and a certain degree of spectral resolution (eight bands), with the number of bands, width, and height being s, W, and H, respectively. HR-PAN has only one band, but its spatial resolution is more than ten times ($\times 12$) that of LR-HSI. For the FHM, its observational model can be expressed as follows:

$$Y = X_M R_P, Z = D_1 X_M \tag{1}$$

where X_M denotes MR-HSI, $R_1 \in \mathbb{R}^{S \times s}$ represents the spectral response function of the satellite sensor's multispectral bands, and $D_1 \in \mathbb{R}^{\mathrm{WH} \times W_P H_P}$ represents the degradation operator between MR-MSI and LR-HSI, which is typically considered to consist of a circular convolutional operator and a downsampling matrix. For the FHP, its observational model can be expressed as follows:

$$P = X_P R_2, X_M = D_2 X_P R_2 \tag{2}$$

where X_P denotes MR-HSI, $R_2 \in \mathbb{R}^{S \times 1}$ represents the spectral response function of the satellite sensor's HR-PAN band, $D \in$

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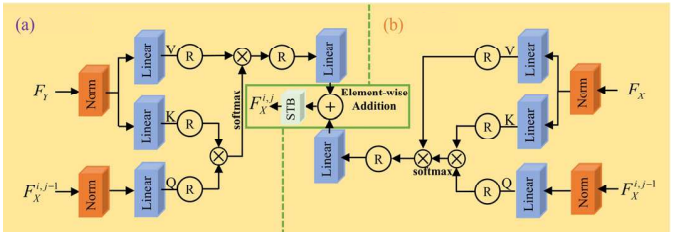


Fig. 2. Architecture of the proposed SSICFB. (a) SACB. (b) SECB.

$\mathbb{R}^{wh\times \mathrm{WH}}$ represents the degradation operator between HR-PAN and MR-MSI.

1) Fusion of LR-HSI and MR-MSI: This article uses a Gaussian convolutional kernel with a size of 8 × 8 and a standard deviation of 2 to obtain DLR-HSI $Z_{\mathrm{Down}} \in \mathbb{R}^{(w/u_1)(h/u_1)\times S}$ by downsampling LR-HSI Z by a factor of u_1. Simulation of LR-MSI $Y_1 \in \mathbb{R}^{wh\times s}$ using the spectral response function of the satellite sensor's multispectral bands to downsample LR-HSI under the spectrum

$$Z_{\mathrm{Down}} = G_{\mathrm{Down}}(Z), Y_1 = R_1(Z) \tag{3}$$

where $G_{\mathrm{Down}}(\cdot)$ represents the Gaussian convolution kernel used and u_1 represents the ratio of spatial resolution between the original hyperspectral and multispectral data. In the first step of training, $Z_{\mathrm{Down}} \in \mathbb{R}^{(w/u_1)(h/u_1)\times S}$ and $Y_1 \in \mathbb{R}^{wh\times s}$ are used as inputs, while the original hyperspectral image $Z \in \mathbb{R}^{wh\times S}$ is used as a reference image for training the dual-input fusion model SSICFNet. Then, map the trained model to a higher dimension, input LR-HSI $Z \in \mathbb{R}^{wh\times S}$ and MR-MSI $M \in \mathbb{R}^{\mathrm{WH}\times s}$ to obtain MR-HSI $X_M \in \mathbb{R}^{\mathrm{WH}\times S}$.

2) Fusion of MR-HSI and HR-PAN: This article used a Gaussian convolution kernel with a size of 8 × 8 and a standard deviation of 2 to obtain DMR-HSI $X_{M\cdot\mathrm{Down}} \in \mathbb{R}^{(W/u_2)(H/u_2)\times S}$ by downsampling MR-HSI X_M. Simulate MR-PAN $P_1 \in \mathbb{R}^{\mathrm{WH}\times 1}$ using the spectral response function of the corresponding satellite sensor's HR-PAN band. DMR-HSI and MR-PAN are used as inputs for training the FHP fusion model, while the MR-HSI generated in the FHM are used as reference images for training

$$X_{W\cdot\mathrm{Down}} = G_{\mathrm{Down}}(X_W), P_1 = R_2(X_W). \tag{4}$$

Then, the trained model is used to fuse MR-HSI and HR-PAN to obtain large-scale hyperspectral fusion images HR-HSI X_P.

B. Overall Framework of SSICFNet

We proposed a novel dual-input fusion framework, the spatial–spectral cross-fusion network (SSICFNet), whose architecture is illustrated in Fig. 1(c). The model employs a feature extraction module to generate spatial features such as edges and textures, spectral features including band correlations, and combined features that refer to new representations generated by intelligently merging spatial details and spectral information from different input images. Subsequently, the extracted features from different depth spatial–spectral cross-fusion groups (SSICFGs) are adaptively fused using trainable weighting parameters. Finally, the fused features are processed through an image reconstruction module to generate HR-HSI. The SSICFG architecture integrates multiple spatial–spectral cross-fusion blocks (SSICFBs, Fig. 2) with a terminal convolutional layer. The SSICFB integrates two novel modules: 1) the SECB, which employs spectral attention to optimize spectral fidelity by effectively extracting spectral features and 2) the SACB, which enhances spatial details through spatial attention mechanisms. The synergistic operation of SACB and SECB modules enables high-quality fusion of hyperspectral and multispectral (or panchromatic) images. The SSICFNet comprehensively integrates the SACB and SECB modules, as well as the advantages of CNN feature extraction, enabling it to demonstrate excellent fusion performance under training methods that follow Wald's principle. The overall implementation process of SSICFNet is as follows.

First, this network uses bilinear interpolation to increase the spatial scale of Z so that its spatial dimension remains consistent with Y

$$Z_{\mathrm{up}} = H_{\mathrm{up}}(Z). \tag{5}$$

Subsequently, Y and Z_{up} are aggregated along the channel dimension to form $D_{cat} \in \mathbb{R}^{\mathrm{WH}\times(S+s)}$, and a 3 × 3 convolutional layer $H_{\mathrm{SF}}(\cdot)$ is used to extract shallow features $F_0 \in \mathbb{R}^{\mathrm{WH}\times C}$

$$F_0 = H_{\mathrm{SF}}\left(\mathrm{Cat}\left(Z_{\mathrm{up}}, Y\right)\right) \tag{6}$$

where $\mathrm{Cat}(\cdot)$ represents a connection operator. In the process of deep feature extraction, two 3 × 3 convolutional layers $H_{\mathrm{spa}}(\cdot)$ and $H_{\mathrm{spe}}(\cdot)$ are used to extract $Z_1 F_{\mathrm{spe}} \in \mathbb{R}^{\mathrm{WH}\times C}$ spectral features and $YF_{\mathrm{spa}} \in \mathbb{R}^{\mathrm{WH}\times C}$ spatial features

$$F_{\mathrm{spa}} = H_{\mathrm{SF1}}(Y), F_{\mathrm{spe}} = H_{\mathrm{SF2}}(Z_1) \tag{7}$$

where C represents the number of feature channels. Then, the features $F_{\mathrm{spa}}, F_{\mathrm{spe}}$, and F_0 are applied to the fusion of spatial and spectral information by K SSICFG

$$F_i = H_{\mathrm{SSICFG}}(F_{i-1}, F_{SPE}, F_{SPA}). \tag{8}$$

Finally, a set of trainable features are used to regulate feature inputs of different depths and fuse them to obtain MR-HSIs X_M

$$X_M = H_{\mathrm{REC}}(F_0 + F_1 \times w_1 + F_2 \times w_1 + \cdots + F_K \times w_K) \tag{9}$$

where $w_1, w_2, \ldots, w_K$ correspond to feature weight factors. $H_{\mathrm{REC}}(\cdot)$ represents the image reconstruction module, which uses a 3 × 3 convolutional layer to complete the task.

C. Spatial–Spectral Cross-Fusion Group

The SSICFG consists of N SSICFB and a convolution layer (see Fig. 2). To extract features from deep neural networks at different depths and alleviate the vanishing gradient problem, SSICFG uses the following deep dense connections [39]:

$$F_i = \mathrm{Conv}\left(\mathrm{Cat}\left(F_0, F_1, \ldots, H_{SSICFB_i}(F_{i-1}, F_{SPE}, F_{SPA})\right)\right). \tag{10}$$

Here, $i = 1, 2, \ldots, K$, and $\mathrm{Conv}(\cdot)$ represents a 3 × 3 convolutional layer, while $\mathrm{Cat}(\cdot)$ represents a connection

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operator. $H_{SSICFB_i}(\cdot)$ represents the ith SSICFB module, while F_i represents the ith fusion feature. Given as input, then

$$F_{i,1} = H_{SSICFB_1}(F_i), F_{i,j} = H_{SSICFB_j}\left(F_{i,j-1}\right) \tag{11}$$

$$F_{i+1} = H_{\mathrm{SF}}\left(H_{SSICFB}\left(F_{i,N}\right)\right) \tag{12}$$

where $j = 1, 2, \ldots, N$, $H_{SSICFB_i}(\cdot)$ represents the jth SSICFB. Then, add a 3×3 convolutional layer before the residual connection $H_{CONV}(\cdot)$. The output of SSICFG can be illustrated as follows:

$$F_{i,\mathrm{out}} = H_{CONV}\left(F_{i,N}\right) + F_i. \tag{13}$$

D. Spatial–Spectral Cross-Fusion Block

The SSICFB consists of an SECB an SACB and a Swin Transformer block (STB) [40]. The specific implementation process is shown in Fig. 2. For the input $F_{i,j-1}, F_X$, and F_Y, first flatten them to $\{P^1, P^2, \ldots, P^M\}$, $\{P_X^1, P_X^2, \ldots, P_X^M\}$ and $\{P_Y^1, P_Y^2, \ldots, P_Y^M\}$, where $P^m, P_X^m, P_Y^m \in \mathbb{R}^{L^2 \times C}$, $m = 1, 2, \ldots, M, M = (\mathrm{HW}/L^2)$. Then, P^m is linearly projected onto *key* $K \in \mathbb{R}^{L^2 \times d}$ and *value* $V \in \mathbb{R}^{L^2 \times d}$. For SECB, P_X^m is linearly projected onto *query* $Q_X \in \mathbb{R}^{L^2 \times d}$. For SACB, P_Y^m is linearly projected onto *query* $Q_Y \in \mathbb{R}^{L^2 \times d}$

$$K = P^m W^K, V = P^m W^V, Q_X = P_X^m W^Q, Q_Y = P_Y^m W^Q. \tag{14}$$

Here $W^Q, W^K, W^V \in \mathbb{R}^{C \times d}$ represents the projection matrix. Then, calculate the attention in each attention head of SECB and SACB

$$CA_X = \mathrm{sofamax}\left(Q_X K^T / \sqrt{d} + B\right) V$$

$$CA_Y = \mathrm{sofamax}\left(Q_Y K^T / \sqrt{d} + B\right) V \tag{15}$$

where B is the learnable relative position encoding. After executing h cross-attention operations in parallel, the results are concatenated to achieve multihead cross-information fusion. The entire process is described as follows:

$$F_{i,j}^X = \mathrm{SECB}\left(\mathrm{LN}\left(F_{i,j-1}\right), \mathrm{LN}\left(F_X\right)\right) \tag{16}$$

$$F_{i,j}^Y = \mathrm{SACB}\left(\mathrm{LN}\left(F_{i,j-1}\right), \mathrm{LN}\left(F_Y\right)\right). \tag{17}$$

The SECB integrates LR-HSI (or MR-HSI) features into joint features through spectral cross-attention computation, whereas the SACB incorporates LR-HSI (or MR-HSI) features via spatial cross-attention mechanisms. The concatenation of spectral and spatial cross-attention outputs achieves comprehensive fusion of spatial–spectral information. After obtaining $F_{i,j}^X$ and $F_{i,j}^Y$, add them together, use the STB to integrate the overall features, and finally integrate all the information through a 3×3 convolutional layer $H_{\mathrm{REC}}(\cdot)$ to reconstruct the fusion image

$$F_{i,j} = H_{\mathrm{STB}}\left(F_{i,j}^X + F_{i,j}^Y\right) \tag{18}$$

where $H_{\mathrm{STB}}(\cdot)$ stands for STB module. By integrating the above modules, this article constructs the spatial–spectral cross-fusion network (SSICFNet).

IV. Experiments

A. Experimental Setting

To validate the fusion performance of the proposed GFHMP framework and SSICFNet dual-input model, we conduct three experiments on both the public simulated Chikusei dataset obtained by an airborne platform and the custom real-world M-ZY1E dataset obtained by a satellite platform: image fusion quality assessment experiments, model performance verification experiments, and reconstruction of hyperspectral image inversion performance verification experiments.

1) Image Fusion Quality Assessment Experiments: This experiment focuses on verifying the fusion image quality of the GFHMP fusion framework on the Chikusei [41] dataset and the M-ZY1E dataset. For quantitative assessment, three metrics are applied to the Chikusei dataset results: peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM), and root mean square error (RMSE). The spatial quality of the fused images is qualitatively evaluated using image detail visualization methods. Two metrics, spectral angle mapper (SAM) and erreur relative globale adimensionnelle de synthèse (ERGAS), were selected to qualitatively assess the spectral quality of the fused images. The spectral fidelity of the fused images was qualitatively assessed through spectral curve comparisons and vegetation index regression analysis. For the M-ZY1E dataset, we evaluate the spatial quality of fusion results through visual comparison with original images, while spectral fidelity is quantitatively analyzed using spectral density curves.

2) Model Performance Verification Experiments: This section presents the performance evaluation of the proposed SSICFNet through comparative experiments and ablation studies. The comparative experiments were performed on both the Chikusei and M-ZY1E datasets, evaluating SSICFNet against five state-of-the-art methods: MSDCNN, PSRT, TFNet, DCTransformer, and MIMFormer. For the Chikusei dataset, spatial quality was quantitatively assessed using three metrics, including PSNR, SSIM, and RMSE. Texture reconstruction performance was assessed through error map analysis between the fused and reference images. The spectral fidelity was quantitatively assessed using the spectral SAM and ERGAS metrics. For the M-ZY1E dataset, the fusion quality was evaluated through visual comparison of the image details of different models. All compared methods were implemented using the parameter settings recommended in their original publications. The ablation studies were conducted on the Chikusei dataset to evaluate module effectiveness through sequential removal. The optimal depth of the model was selected by gradually increasing the number of SSICFG.

3) Reconstruction of Hyperspectral Image Inversion Performance Verification Experiments: This experiment systematically evaluates the feasibility and advantages of GFHMP and SSICFNet reconstructed imagery for quantitative inversion applications. We employed an exhaustive computational method to derive spectral indices and extract characteristic information across all available wavelength bands. Two methods, removal of highly correlated features and recursive feature elimination with cross-validation (RFECV), were used

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TABLE I
FUSION RESULTS OF GFHMP FRAMEWORK ON THE CHIKUSEI DATASET

Steps	PSNR (↑)	SSIM (↑)	SAM (↓)	ERGAS (↓)	RMSE (↓)
FHM	43.94	0.9933	1.729	0.979	0.00485
FHP	42.94	0.9643	3.754	2.176	0.01080

to screen features and reduce the dimensionality of the data, completing the creation of the optimal inversion dataset. SHAP were employed to quantify both global and local feature importance, elucidating the inverse modeling mechanism.

In our implementation, we set the number of SSICFNet modules to $k = 3$, trained the model for 100 epochs on an NVIDIA Quadro GV100 GPU with an initial learning rate of 0.0001, and optimized the parameters by minimizing the $L1$ loss.

B. Datasets

1) Chikusei Dataset: A publicly available simulated dataset was utilized in this study. The airborne hyperspectral dataset was acquired over Chikusei City, Ibaraki Prefecture, Japan (July 29, 2014), with dimensions of 2517 × 2335 × 128. The data covers the spectral range of 0.360–1.018 μm at 2.5-m spatial resolution. A central 2304 × 2304 × 128 pixel subset of the hyperspectral imagery was selected as the reference HR-HSI (Ref-HR-HSI, 2.5 m). To simulate satellite observations, the ZY1E spectral response function was applied to generate high-resolution multispectral (HR-MSI, 2.5 m) and panchromatic (HR-PAN, 2.5 m) imagery. Subsequently, HR-HSI and HR-MSI were spatially downsampled using an 8 × 8 Gaussian convolution kernel ($\sigma = 2$) to generate low-resolution hyperspectral (LR-HSI, 30 m) and medium-resolution multispectral (MR-MSI, 10 m) images, respectively. In the FHM framework, LR-HSI undergoes further downsampling to generate a downsampled low-resolution hyperspectral image (DLR-HSI, 90 m) and a low-resolution multispectral image (LR-MSI, 30 m) for training input. Training samples consist of randomly extracted 16 × 16 × 128 DLR-HSI and 48 × 48 × 8 LR-MSI patches, with 48 × 48 × 128 LR-HSI patches serving as reference data. In the FHP, MR-HSI and HR-PAN outputs from FHM fusion were downsampled (×4) using a Gaussian convolution kernel, generating a downsampled MR-HSI (DMR-HSI, 40 m) and medium-resolution Panchromatic image (MR-PAN, 10 m). During training, random 16 × 16 × 128 DMR-HSI and 64 × 64 × 1 MR-PAN patches were extracted as training samples, with corresponding 64 × 64 × 128 MR-HSI patches serving as reference data.

2) M-ZY1E Dataset: A custom satellite dataset was utilized in this study. The ZY1E satellite, China's first civilian hyperspectral mission, simultaneously acquires panchromatic imagery at 2.5-m resolution, multispectral data at 10-m resolution, and hyperspectral observations at 30-m resolution. Its hyperspectral sensor covers 0.400–2.500 μm through 76 VNIR and 90 SWIR bands. We acquired LR-HSI, MR-MSI, and HR-PAN over the Qinzhou Coastal Mangrove Reserve on October 7, 2021, with sizes of 645 × 383 × 158, 1931 × 1147 × 8, and 7728 × 4586 × 1, respectively. Following preprocessing, including radiometric calibration, atmospheric correction, georeferencing, and cropping, the raw data were processed into six hyperspectral, multispectral, and panchromatic datasets (M_1, M_2, …, M_6, as shown in Table I of Supplementary Material). The first dataset (M_1) encompasses the coastal mangrove forests of Qinzhou Coastal Mangrove Reserve. To maintain fused image integrity, the panchromatic and multispectral images in each M-ZY1E dataset group were collected with a slightly larger spatial extent than the corresponding hyperspectral images. Owing to the low signal-to-noise ratio (SNR) in specific spectral bands of the ZY1E data, only the first 76 bands of LR-HSI were utilized for fusion experiments. During training dataset construction, random patches are first extracted from the LR-HSI, followed by acquisition of corresponding MR-MSI and HR-PAN regions using geographic coordinates. In the FHM, LR-HSI and MR-MSI were downsampled (×3) using an 8 × 8 Gaussian convolution kernel ($\sigma = 2$) to generate DLR-HSI and LR-MSI as training inputs. During training, random DLR-HSI patches of size 8 × 8 × 128 and LR-MSI patches of size 24 × 24 × 8 were extracted as training samples, with matching LR-HSI reference patches of size 24 × 24 × 76. In the FHP, MR-HSI and HR-PAN were downsampled (×4) using an 8 × 8 Gaussian kernel ($\sigma = 2$) for fusion, generating DMR-HSI and MR-PAN as training inputs. Random 8 × 8 × 128 MR-HSI and 32 × 32 × 1 HR-PAN patches were extracted as training samples, with corresponding 32 × 32 × 76 MR-HSI patches serving as reference data.

C. Quality Assessment of Fusion Images

The fusion performance metrics of the GFHMP framework on the Chikusei dataset are quantitatively presented in Table I. The best-performing metrics are highlighted in bold, while the second-best are underlined. The FHM stage achieves a PSNR = 43.75 dB and SSIM = 0.9930, demonstrating that the GFHMP framework effectively preserves MR-MSI structural and textural information during initial fusion while reconstructing high-quality images. The fused image demonstrates excellent spectral preservation, with SAM = 1.756°. With ERGAS = 1.000 and RMSE = 0.00496, the proposed method achieves superior global error control, significantly enhancing reconstruction quality. The FHP method achieved PSNR = 42.94 dB and SSIM = 0.9643, representing reductions of 1.00 dB and 0.0290, respectively, relative to FHM fusion. These results indicate that while FHP demonstrates marginally reduced spatial fidelity compared to the FHM, it retains effective performance in preserving spatial details and texture information. Spectral evaluation results indicate that while FHP achieved a lower SAM = 3.754° compared

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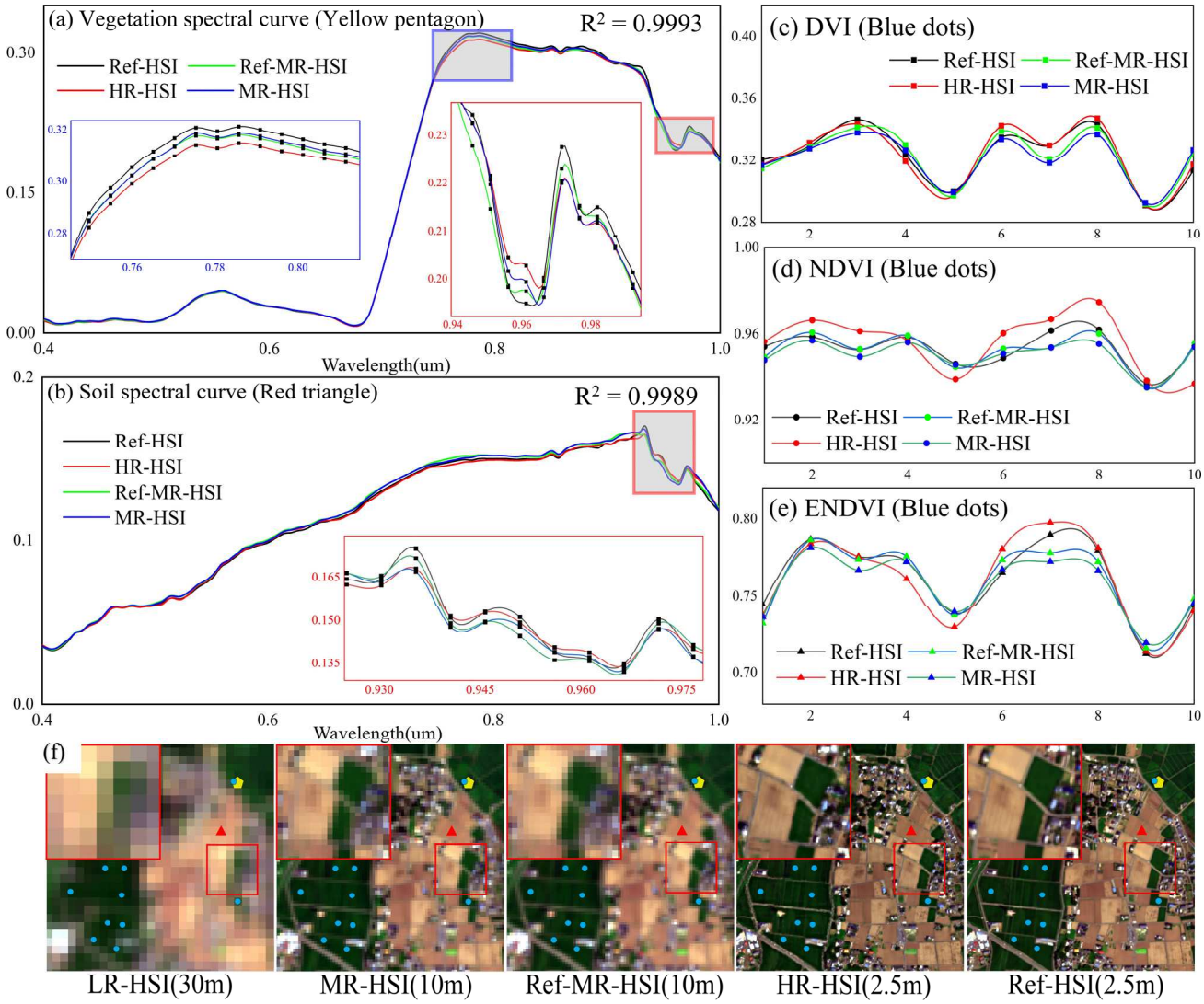


Fig. 3. Visual evaluation for the fusion result of GFHMP on the Chikusei dataset (red: 60 and green: 40, blue: 20) and comparative analysis of spectral curves. (a) Comparison of vegetation spectral curves (yellow pentagon). (b) Comparison of soil spectral curves (red triangle). (c) Comparative analysis of DVI vegetation index (blue dots). (d) Comparative analysis of the NDVI vegetation index (blue dots). (e) Comparative analysis of VLI vegetation index (blue dots). (f) Visual evaluation.

with FHM, it still maintains high spectral fidelity. While a slight spectral distortion occurred during MR-HSI fusion, the obtained sub-5° SAM value verifies that the HR-HSI retains high spectral fidelity. Fig. 3 presents the spectral evaluation and visual comparison results for the fused Chikusei dataset images, including vegetation and soil spectral curves along with vegetation index comparisons among LR-HSI, MR-HSI, HR-HSI, and Ref-MR-HSI, accompanied by visualizations of the fused image details.

Fig. 3(a) shows the vegetation spectral curves of the vegetation points (marked by a yellow pentagonal point) in Fig. 3(f). The spectral curves of all four images demonstrate complete consistency across the 0.4–0.75-μm wavelength range, with the reconstructed image perfectly preserving the vegetation spectral characteristics. The spectral curves exhibit significant variations in the 0.75–0.85-and 0.94–0.99-μm wavelength ranges, where spectral preservation proves more challenging. Nevertheless, the four image curves maintain remarkably consistent spectral characteristics across these spectral regions. Fig. 3(b) presents the spectral curves corresponding to the soil sampling locations (marked by a red triangular point) in Fig. 3(f). The spectral curves of the four images in the 0.4–0.92- and 0.98–1.00-μm ranges are basically overlapping, indicating that the spectral fidelity of the reconstructed images is very high. Within the range of 0.92–0.98 μm, the spectral curves exhibit differences compared to other wavelength ranges, but still maintain an extremely high degree of fit. Furthermore, the R^2 = 0.9993 (vegetation) and R^2 = 0.9989 (soil) between HR-HSI and Ref-HSI in Fig. 3(a) demonstrate both the effectiveness of the GFHMP method and the excellent spectral fidelity of the reconstructed imagery. Three vegetation indices on ten vegetation points (marked by blue dots) in Fig. 3(f) are shown in Fig. 3(c)–(e): difference vegetation index (DVI), normalized DVI (NDVI), and enhanced NDVI (ENDVI), respectively. The quality of vegetation fusion was further validated by comparative analysis. Fig. 3(f) displays

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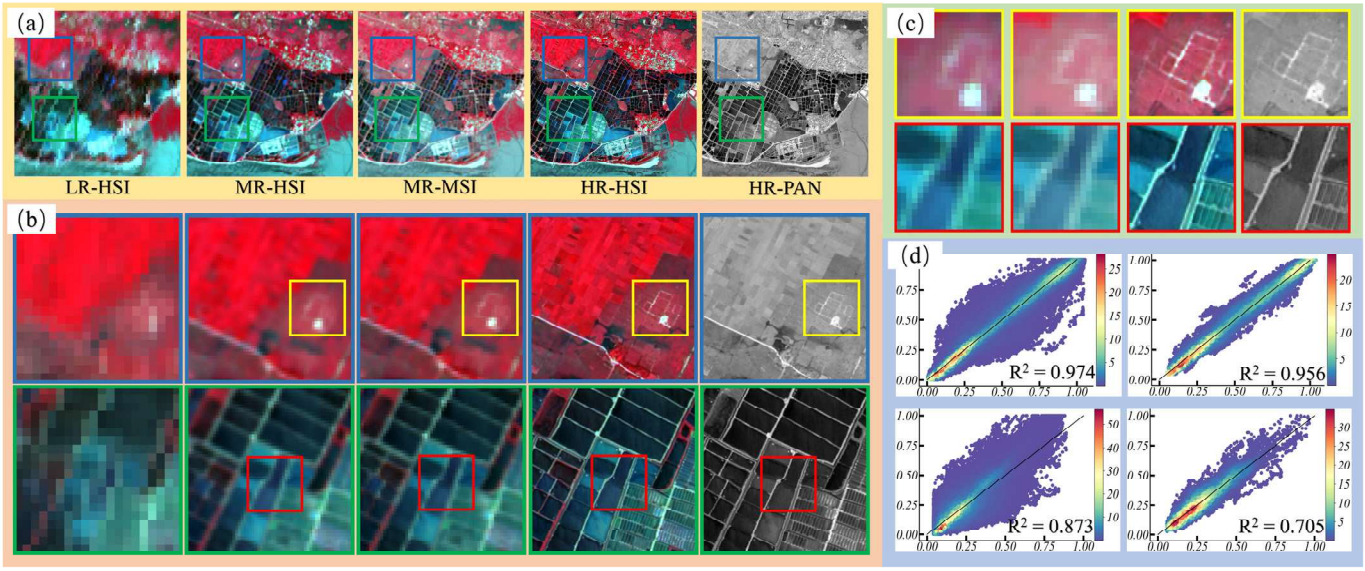


Fig. 4. Visual evaluation and spectrum evaluation for the fusion result of GFHMP on the M-ZY1E dataset. (a) Wide-area visual evaluation. (b) Zoomed-in views of the blue and green boxes in (a). (c) Zoomed-in views of the blue and green boxes in (b). (d) Spectral density scatter of LR-HSI versus MR-HSI and MR-HSI versus HR-HSI in (b).

TABLE II
COMPARISON OF FHM FUSION RESULTS OF DIFFERENT ALGORITHMS ON THE CHIKUSEI DATASET

Method	PSNR (↑)	SSIM (↑)	SAM (↓)	ERGAS (↓)	RMSE(↓)
MSDCNN	38.50	0.9863	3.225	1.832	0.00908
PSRT	33.25	0.9889	5.877	3.350	0.01661
TFNet	43.25	0.9905	1.868	1.059	0.00525
DCTransformer	42.36	0.9919	2.043	1.173	0.00581
MIMFormer	42.12	0.9902	2.093	1.206	0.00598
Ours	**43.94**	**0.9933**	**1.729**	**0.979**	**0.00485**

the original image (LR-HSI), the merged images (MR-HSI, HR-SHI), and the reference images (MR-Ref-HSI, HR-HSI) in a visual comparison, with details of each image displayed in the enlarged view within the red frame. As the resolution increases in the three images, the details of the terrain become clearer, while the spectral information is also well preserved from a visual perspective. The MR-HSI and Ref-MR-HSI images in Fig. 3 are basically consistent, indicating that the FHM fusion results are excellent. This can be attributed to the difference in scale between the fused images is small ($\times 3$), and MR-MSI itself has relatively rich spectral information. Comparative analysis with Ref-HSI demonstrates that HR-HSI maintains nearly identical spatial textures and spectral characteristics, achieving excellent overall reconstruction performance. However, minor spectral distortions are observed along some feature boundaries, primarily due to the relatively large fusion scale factor ($\times 4$) of FHP and the limited spectral information in the single-band HR-PAN data. Notably, the spectral characteristics of vegetation and soil are exceptionally well preserved in localized regions.

Fig. 4 presents the visual comparison results on the M-ZY1E dataset. Hyperspectral images are visualized using a combination of red: 51, green: 30, and blue: 19 bands, while multispectral images are visualized using a combination of red: 4, green: 3, and blue: 2 bands. The original images (LR-HSI, MR-MSI, and HR-PAN) and fused images (MR-HSI and HR-HSI) of three different regions are presented in Fig. 4, respectively. Fig. 4 demonstrates that MR-HSI maintains high spatial consistency with the original MR-MSI in both texture and detail preservation, confirming the effectiveness of the proposed SSICFNet for FHM. However, subtle spectral discrepancies exist between MR-HSI and MR-MSI, primarily attributable to the inherent differences between hyperspectral and multispectral imaging systems. The GFHMP method accomplished large-scale ($\times 12$) fusion of ZY1E satellite hyperspectral data via a two-step fusion process. The HR-HSI exhibits nearly identical spatial details and texture information to the original panchromatic image, demonstrating effective extraction of high-resolution spatial features from the HR-PAN data. Visual comparison of spectral information fusion quality among HR-HSI, MR-MSI, and LR-HSI demonstrates that the reconstructed images maintain excellent spectral fidelity, preserving comprehensive spectral characteristics. Both spectral density scatter plots show R^2

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TABLE III
COMPARISON OF FHP FUSION RESULTS OF DIFFERENT ALGORITHMS ON THE CHIKUSEI DATASET

Method	PSNR (↑)	SSIM (↑)	SAM (↓)	ERGAS (↓)	RMSE (↓)
MSDCNN	39.28	0.9387	5.727	3.314	0.01643
PSRT	32.42	0.9425	12.480	7.303	0.03620
TFNet	<u>41.75</u>	0.9538	4.307	<u>2.496</u>	<u>0.01238</u>
DCTransformer	41.64	<u>0.9576</u>	<u>4.251</u>	2.527	0.01253
MIMFormer	41.64	0.9545	4.303	2.527	0.01253
Ours	**42.94**	**0.9643**	**3.754**	**2.176**	**0.01080**

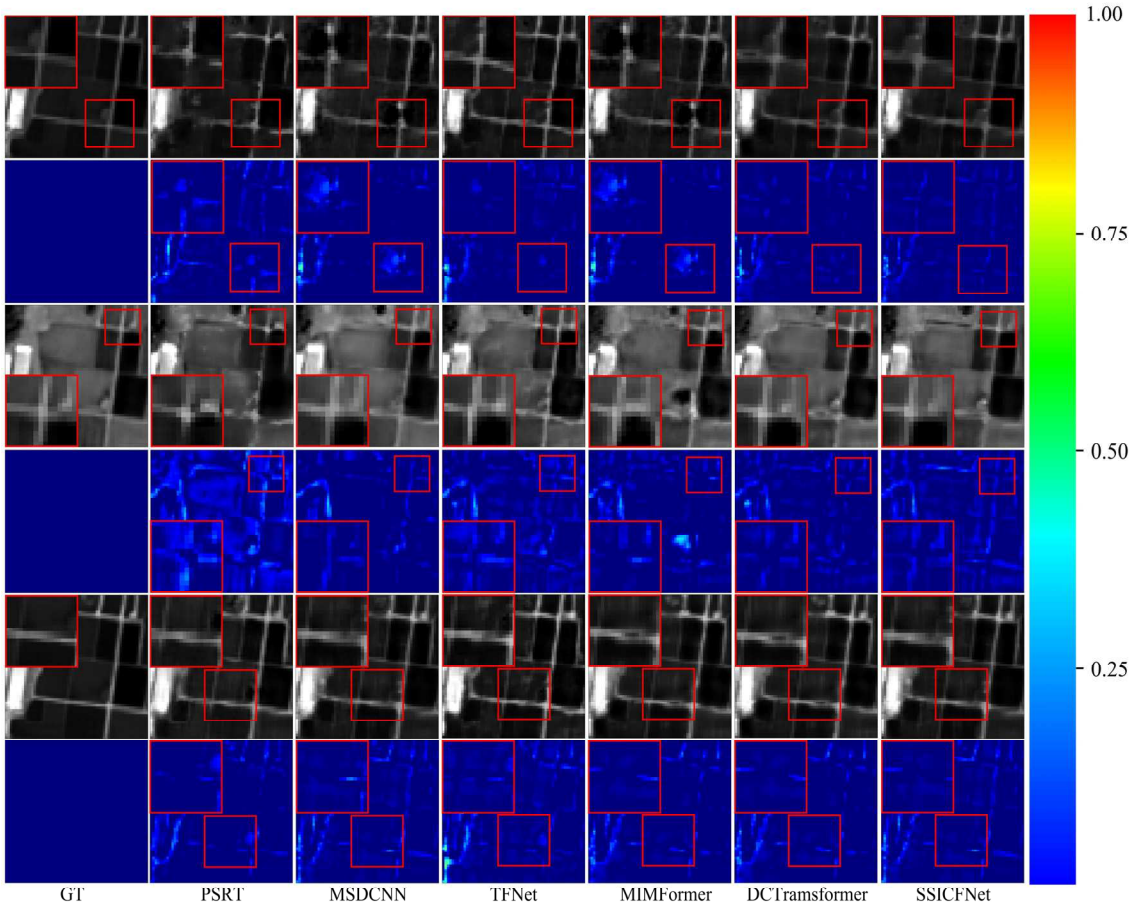


Fig. 5. Qualitative results of FHP derived from the Chikusei dataset at 20th, 40th, and 60th bands are displayed. The odd-numbered row presents the ground truth and the images reconstructed by all compared methods for the 20th, 40th, and 60th bands, respectively. The even-numbered row depicts the error images between the fused and the ground truth.

values exceeding 0.7. This fully demonstrates that the GFHMP fusion method can effectively achieve large-scale fusion of satellite hyperspectral images, generating hyperspectral images with high spatial resolution.

D. Model Performance Verification

In this section, a series of experiments were conducted on the Chikusei and M-ZY1E datasets to verify the superiority of the proposed network. This study compares the proposed SSICFNet with six state-of-the-art methods, including two CNN-based methods (MSDCNN and TFNet) and four recently developed Transformer-based approaches (PSRT, MIMFormer, DCTransformer, and Fusformer). Through comparative experiments and an ablation study, the ability of spatial and spectral information extraction and fusion of the SSICFNet model is comprehensively evaluated.

1) Comparative Experiments on the Chikusei Dataset: Tables II and III present the FHM and FHP fusion results for all comparison models on the Chikusei dataset, evaluated using five metrics: PSNR, SSIM, SAM, ERGAS, and RMSE. The best-performing metrics are highlighted in bold, while the second-best are underlined. All fused images exhibit high

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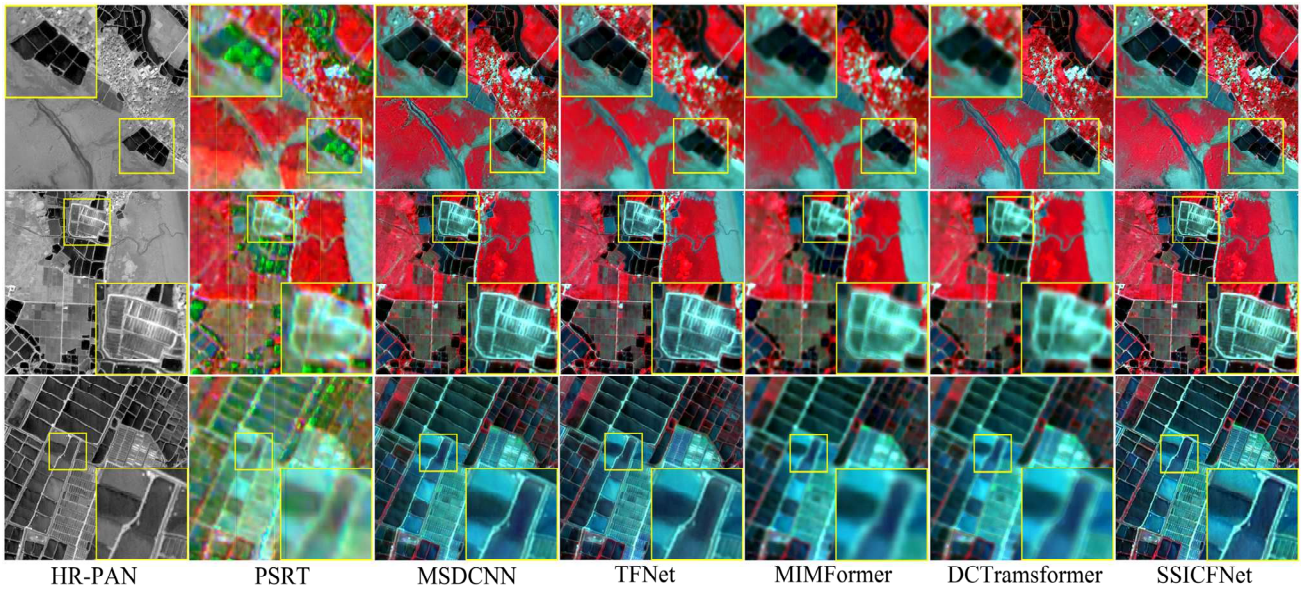


Fig. 6. Visual comparison for the fusion results of FHM on the M-ZY1E dataset by standard false color composite (red: 51, green: 30, and blue: 19).

TABLE IV
INFLUENCE OF THE NUMBER OF SSICFG OF THE NETWORK

SSICFG number	Step	PSNR (↑)	SSIM (↑)	SAM (↓)	ERGAS (↓)	RMSE (↓)
2	FH-M	43.55	0.9930	**1.706**	1.024	0.00508
	FH-P	42.80	0.9637	**3.728**	2.211	0.01096
3	FH-M	**43.94**	**0.9933**	1.729	**0.979**	**0.00485**
	FH-P	**42.94**	**0.9643**	3.754	**2.176**	**0.01080**
4	FH-M	43.81	0.9930	1.754	0.994	0.00493
	FH-P	42.84	0.9642	3.792	2.200	0.01091

similarity to the reference images in structure, texture, and spectral characteristics. The proposed SSICFNet outperforms existing methods in both FHM and FHP tasks. For the FHM, the fusion accuracy is significantly higher than that of the FHP since the training and fusion images are acquired from actual sensors. In the FHM, SSICFNet achieves improvements of 0.69 dB in PSNR and 0.0014 in SSIM over the suboptimal method, while also demonstrating significant enhancements in SAM, ERGAS, and RMSE metrics. In the FHP, SSICFNet improved the PSNR metric by 1.19 dB and the SSIM metric by 0.0067. These experimental results fully demonstrate the effectiveness and superiority of SSICFNet in the field of hyperspectral image fusion.

In Fig. 5 and Fig. 1 of Supplementary Material, the visualizations show the HR-HSI and MR-HSI generated by various competing methods on the Chikusei dataset. For FHM fusion, all models demonstrated competent fusion performance. However, the PSRT and MIMFormer methods introduced significant artifacts in the FHP. This discrepancy primarily stems from the use of MR-HSI (generated by the first-step algorithm) in the second-step FHP training process, where inherent spectral and spatial deviations exist between the MR-HSI and real images. Although other methods have achieved satisfactory fusion results, they still exhibit some artifacts and spectral differences. This phenomenon is mainly due to the inherent limitations of the training method based on Wald's principle when mapping from low dimensions to high dimensions. Compared with other methods, the SSICFNet proposed in this article can reduce these differences. Fig. 5 shows that the regions with large differences are mainly concentrated in the building and road regions, while in the vegetated regions, all methods achieve good reconstruction results. This indicates that GFHMP can better preserve spatial and spectral information in large natural areas.

2) Comparative Experiments on the M-ZY1E Dataset: Fig. 6 and Fig. 2 of Supplementary Material present the fusion results of the six models on the M-ZY1E dataset. The merged images exhibit high consistency with the MR-MSI reference data in terms of geometric structure and texture details across all landforms. This demonstrates that all network models can effectively capture spatial features from MR-MSI in the FHM task. This occurs because the spatial resolution ratio between hyperspectral and multispectral images in the FHM task is relatively small (only $\times 3$), and the eight spectral bands of MR-MSI provide substantial spatial information. Regarding spectral information fusion, only the MSDCNN-generated

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TABLE V
ABLATION STUDY: SSICFNET ON THE FHM

SECB	SACB	STB	Trainable parameters	Dense connections	PSRT (↑)	SSIM (↑)	SAM (↓)	ERGAS (↓)	RMSE (↓)
×	√	√	√	√	41.45	0.9885	2.217	1.304	0.00647
√	×	√	√	√	43.81	0.9932	1.756	0.993	0.00492
√	√	×	√	√	43.75	0.9930	1.767	1.000	0.00496
√	√	√	×	√	43.47	0.9925	1.789	1.032	0.00512
√	√	√	√	×	43.83	0.9931	1.754	0.992	0.00492
√	√	√	√	√	**43.94**	**0.9933**	**1.729**	**0.979**	**0.00485**

TABLE VI
ABLATION STUDY: SSICFNET ON THE FHP

SECB	SACB	STB	Trainable parameters	Dense connections	PSRT (↑)	SSIM (↑)	SAM (↓)	ERGAS (↓)	RMSE (↓)
×	√	√	√	√	41.95	0.9524	4.185	2.438	0.01201
√	×	√	√	√	42.76	0.9634	3.830	2.221	0.01101
√	√	×	√	√	42.67	0.9628	3.965	2.246	0.01113
√	√	√	×	√	42.40	0.9605	3.972	2.314	0.01147
√	√	√	√	×	42.69	0.9628	3.860	2.239	0.01110
√	√	√	√	√	**42.94**	**0.9643**	**3.754**	**2.176**	**0.01080**

fused images exhibit minor spectral deviations compared with other methods. This difference indicates that MSDCNN exhibits marginally inferior spectral feature extraction performance compared to alternative network architectures. The other methods are all capable of generating fusion images with high spatial and spectral fidelity, fully demonstrating the effectiveness of the proposed GFHMP framework. The PSRT network demonstrated suboptimal performance in FHP fusion, with significant spectral distortion observed in most water bodies. This suggests limited capability in preserving the original hyperspectral characteristics during panchromatic fusion. Both MIMFormer and DCTformer networks exhibited spectral blurring effects at object boundaries. In contrast, the proposed SSICFNet achieves superior spectral reconstruction, maintaining clear spectral details and sharp edge differentiation, which indicates exceptional spectral fusion fidelity. In terms of spatial fusion, the three networks PSRT, MIMFormer, and DCTramsformer performed poorly in terms of feature edge fusion results, exhibiting severe edge blurring issues compared with other images. The overall fusion result of MSDCNN and TFNet spatial texture is excellent, but the two images have artifacts in areas such as roads and bridges. In contrast, the fusion results from SSICFNet feature extremely fine spatial texture and detail, with virtually no spatial artifacts or other issues in the reconstructed images. The above results show that SSICFNet has extremely high fusion benefits in FHP fusion tasks and prove the strong spatial and spectral fusion performance of the network.

3) Ablation Study: To validate the contribution of individual modules in SSICFNet, we perform comprehensive ablation studies on the Chikusei dataset. All models are trained for 100 epochs under identical configurations to ensure fair comparison.

Table IV presents the impact of SSICFG quantity on GFHMP fusion performance. Experimental results demonstrate that configurations with three SSICFG modules achieve superior performance in both FHM and FHP, outperforming both two- and four-module variants across all evaluated metrics (PSNR, SSIM, ERGAS, and RMSE). The reason for this phenomenon is that as the number of SSICFG increases, the network tends to generate low-resolution features, which may become noise when fusing high-resolution images, thereby reducing image quality. On the other hand, when the number of SSICFG is small, the network cannot fully extract and fuse the spatial and spectral detail information of the input images. Therefore, in the SSICFNet model ultimately adopted in this article, the number of SSICFG is set to three to achieve optimal fusion performance.

Tables V and VI present the respective impacts of the SECB, SACB, STB, trainable parameters, and dense connectivity on FHM and FHP fusion performance. The best-performing metrics are highlighted in bold, while the worst-performing are underlined. This study systematically evaluated the specific contributions of each module to the fusion performance of FHM and FHP by gradually removing each module from the model. In both tables, the symbol "×" indicates that the corresponding module has been removed. When removing SEB, the performance of the model is significantly decreased. The PSRT values of FHM and FHP fusion images are reduced to 41.45 and 41.95 dB, respectively, with a decrease of 2.49 and 0.99 dB. This result shows that the SEB module plays a key role in extracting the spectral information of hyperspectral

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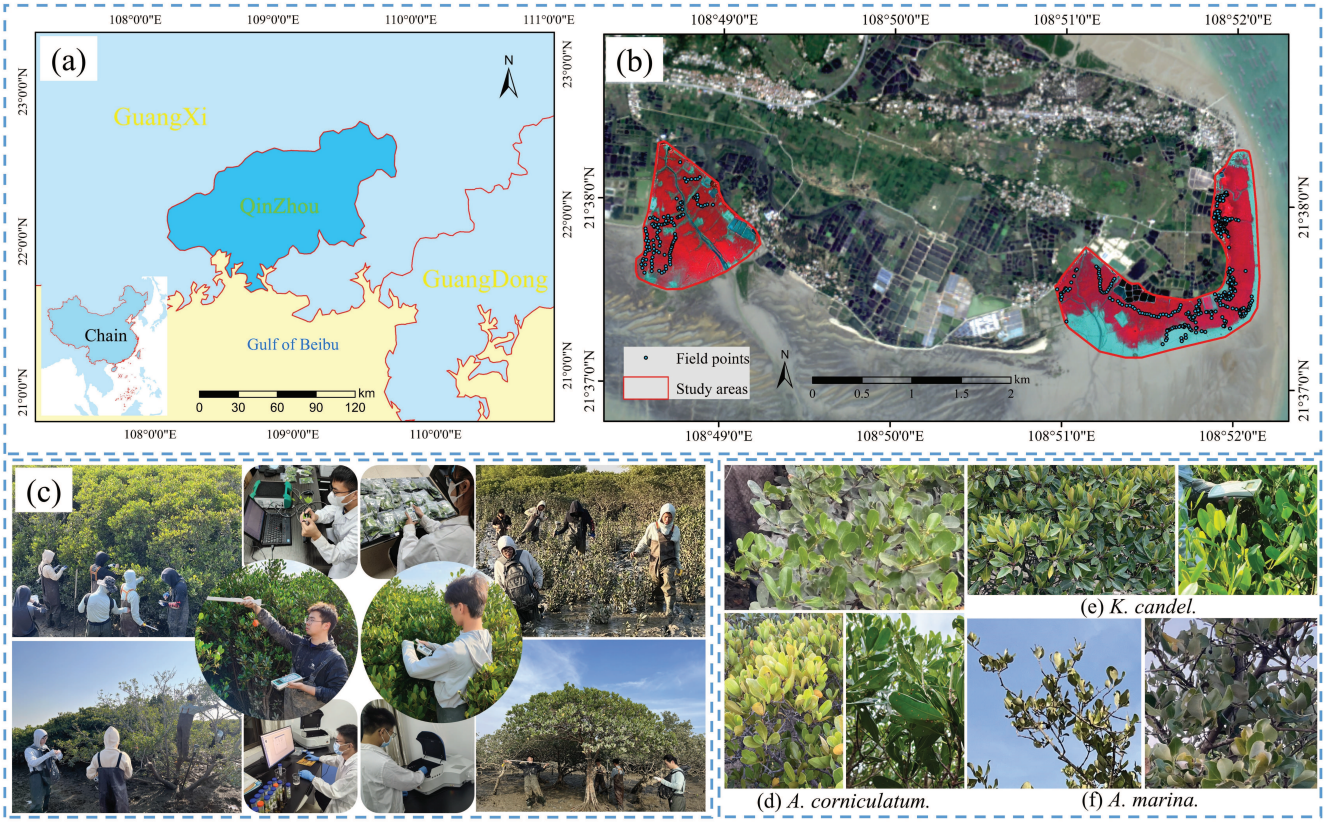


Fig. 7. Geographic location and field sampling in the Beibu Gulf. (a) Geographical location of the QinZhou. (b) Mangrove species. (c) Field collection and laboratory measurement of LCC and LAI. (d) *A. corniculatum.* (e) *K. candel*. (f) *A. marina*.

images and greatly enhances the quality of image fusion. Second, the trainable weight mechanism introduced in this article has a significant impact on the overall performance of the network, which can dynamically adjust the input of different depth residual modules, thereby effectively balancing the fusion of multilevel features. In addition, other modules also played a positive role in improving network performance, achieving higher quality image output in both FHM and FHP tasks.

E. Validation Experiments of Reconstructed Hyperspectral Image on Estimating Mangrove CCC

1) Research Area and Field Data: Mangroves are one of the largest carbon sink ecosystems in the world [42]. The Qinzhou Coastal Mangrove Nature Reserve is located in the subtropical zone with an average annual temperature of 22 °C, which provides an ideal growth environment for mangroves. There are many typical mangrove trees here, such as *Aegicers corniculatum*, *Kandelia candel*, and *Avicennia marina*. CCC is an important indicator for monitoring the health of mangrove populations, and accurate inversion of CCC is of great significance for the protection and monitoring of mangroves. This article takes the coastal mangrove forest reserve in Qinzhou, Guangxi, as the study area (see Fig. 7) and combines field measurement data to conduct a reverse analysis of the chlorophyll content in the canopy of mangrove vegetation in this area. A high-precision chlorophyll inversion model was established by combining multispatial resolution hyperspectral images with field data, providing a scientific basis for assessing the health of mangrove ecosystems.

The field data used in this article includes field measurement data and high-resolution images by drones. This article uses mangrove leaf chlorophyll content (LCC) and leaf area index (LAI) to calculate CCC using the following formula:

$$\mathrm{CCC} = \mathrm{LCC} \times \mathrm{LAI}. \tag{19}$$

LCC and LAI field data were collected from March 30 to April 6, 2021, in the coastal mangrove nature reserve in Qinzhou, Guangxi. At each sampling location, LCC was determined through 20 replicate measurements using a SPAD-502 Plus chlorophyll meter, with the final value representing the arithmetic mean of all measurements. LAI was measured using a handheld LAI-2200C plant canopy analyzer, with each sampling point co-located with corresponding chlorophyll content measurement sites. During the measurement, the geographical position of each sampling point was recorded using a Hi-Target V90 RTK (positioning accuracy 0.25 cm + 1 ppm). A total of 332 samples were collected in this article.

2) Multispatial Resolution Hyperspectral Inversion Scheme Setting: In the experiment, three different spatial resolution images (HR-HSI, MR-HSI, and LR-HSI) were used in combination to invert the chlorophyll content of mangrove canopy, and the following four inversion schemes were setup to explore the benefits of different images in mangrove inversion.

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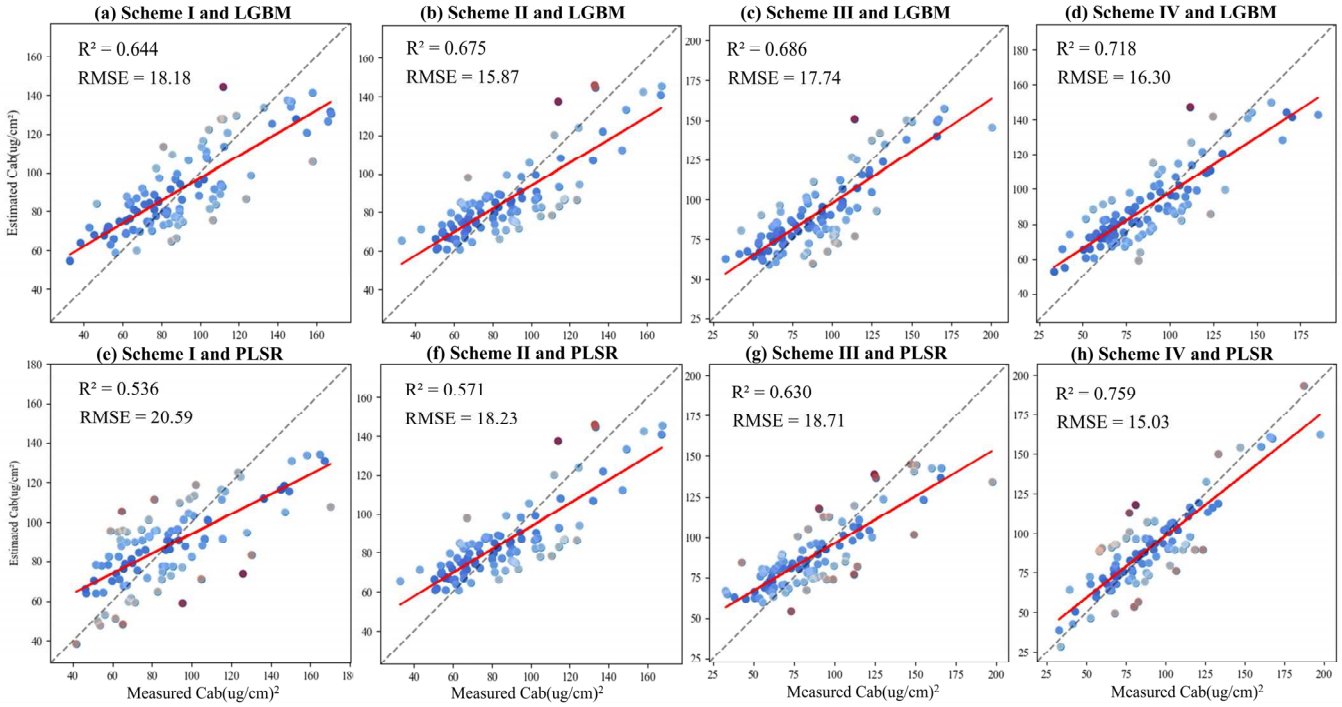


Fig. 8. Estimation accuracy of CCC content in mangroves based on four schemes. (a) CCC inversion is based on both Scheme I and LGBM. (b) CCC inversion is based on both Scheme II and LGBM. (c) CCC inversion is based on both Scheme III and LGBM. (d) CCC inversion is based on both Scheme IV and LGBM. (e) CCC inversion is based on both Scheme I and PLSR. (f) CCC inversion is based on both Scheme II and PLSR. (g) CCC inversion is based on both Scheme III and PLSR. (h) CCC inversion is based on both Scheme IV and PLSR.

TABLE VII
SPECTRAL FEATURE AND TEXTURE FEATURES

Spectral features	Calculation formula	Number of features
$DSI_{(i,j)}$	$DSI = R_i - R_j$	435
$RSI_{(i,j)}$	$RSI = R_i / R_j$	435
$NDSI_{(i,j)}$	$NDSI = (R_i - R_j)/(R_i + R_j)$	435
$NLI_{(i,j)}$	$NLI = (R_i^2 - R_j)/(R_i^2 + R_j)$	435
Texture features	Feature name	
	Mean, Variance, Homogeneity, Contrast, Dissimilarity, Entropy, Second Moment and Correlation	608

Scheme I: The original hyperspectral image (LR-HSI) was used to invert the chlorophyll content of the mangrove canopy, and investigate the effectiveness of LR-HSI for mangrove canopy chlorophyll inversion.

Scheme II: The FHM fusion image (MR-HSI) was used to inversion the chlorophyll content of mangrove canopy, and the efficiency improvement of HSI-MSI fusion for mangrove inversion was explored by comparing with the inversion results of the original image.

Scheme III: The FHP fused image (HR-HSI) is used to invert the chlorophyll content of the mangrove forest canopy. By comparing with the results of the previous two schemes, the advantages of the GFHMP method in inversion are proved.

Scheme IV: The spatial and spectral features obtained by the first three kinds of images are used to jointly inversion the chlorophyll content of the mangrove forest canopy, making full use of image information with different resolutions.

In each scheme, the same method is used to extract the spectral and spatial features of the image. A machine learning model, light gradient boosting machine (LGBM), and a linear regression model, partial least squares regression (PLSR), were used for inversion.

3) Spatial–Spectral Feature Extraction and Selection: Three types of features are used in each scheme for inversion studies. Spectral features: Fine spectral differences between different bands are obtained by calculating composite spectral indices. Texture features: spatial detail information is obtained by performing operations between pixels in a single band. Band features: The top 30 bands with the highest correlation are extracted for inversion studies by calculating the correlation between the original bands and the measured CCC.

a) Spectral feature extraction: Extract the top 30 bands correlated with the CCC actual measurement data from the three images. In order to fully utilize the spectral feature information contained in highly correlated bands, a systematic spectral index construction method is used to comprehensively explore the combination relationships between highly correlated bands, providing sufficient information input for subsequent inversion of chlorophyll content in mangrove canopy layers. We calculate the following four composite spectral indices: differential spectral index (DSI), ratio spectral index (RSI), normalized difference spectral index (NDSI), and nonlinear index (NLI). A total of 1740 spectral features were identified for each image (see Table VII). In Table VII, R_i

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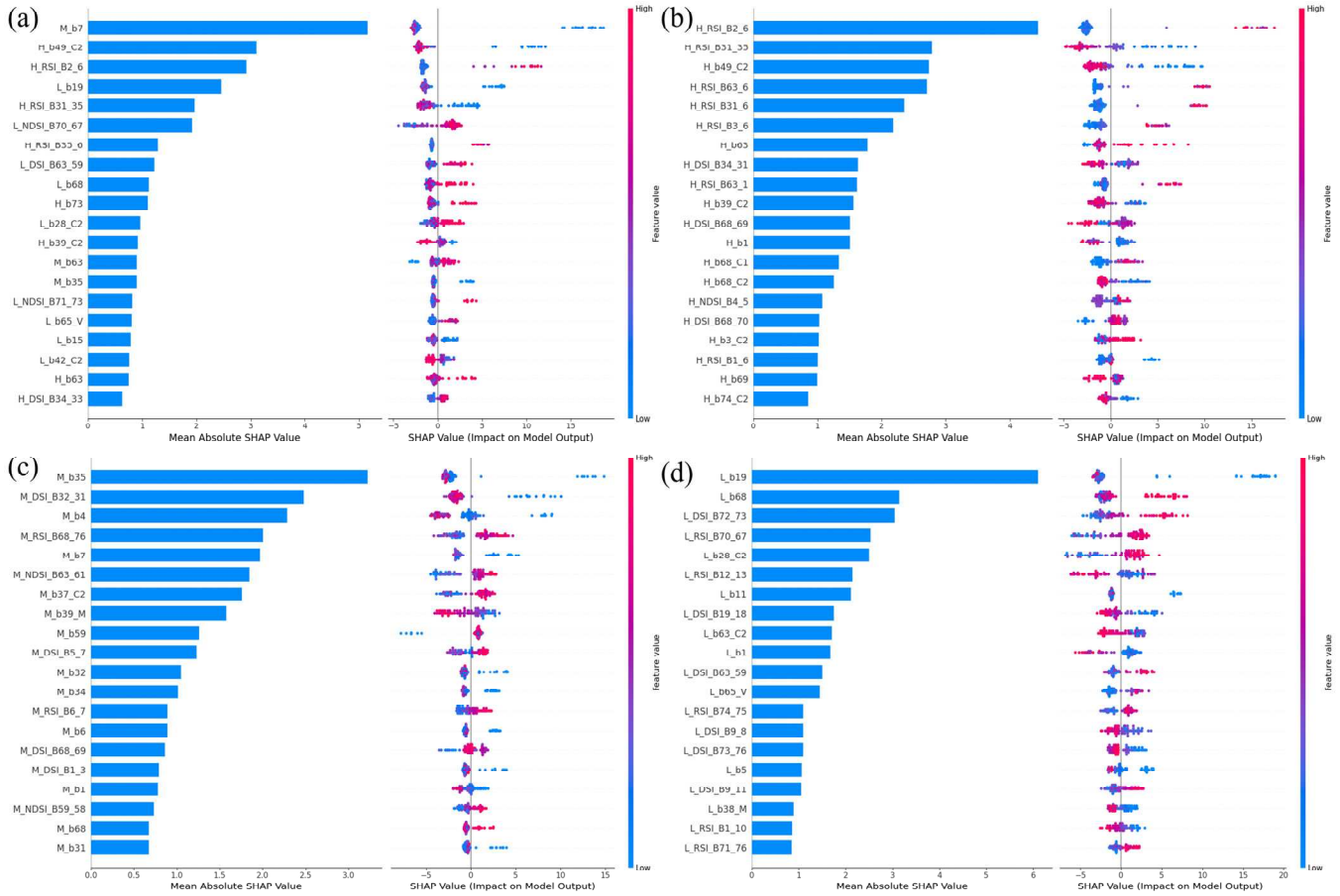


Fig. 9. Global and local interpretation of SHAP values for various input features. (a) Scheme IV. (b) Scheme III. (c) Scheme II. (d) Scheme I.

and R_j represent any two bands in the set of highly correlated bands and satisfy $i \neq j$.

b) Texture feature extraction: We used the co-occurrence measures function of ENVI 5.6 software to extract eight texture features for each band of hyperspectral images (see Table VII), totaling 608 features for each image.

c) Spatial and spectral feature selection: HR-HSI, MR-HSI, and LR-HSI were inverted separately (Schemes I–III) using high correlation to eliminate 1948 redundant features, and retain 200 features to achieve data dimensionality reduction. Combined image inversion (Scheme IV) eliminated 6844 bands with high correlation among the 7044 features of the three images, and retained 200 features to achieve data dimensionality reduction. The RFECV method recursively reduces the feature set size and performs importance ranking, ultimately selecting the optimal number of features through cross validation. When the number of features reached 40, the RFECV method achieved stable cross-validation results with consistently high accuracy. Ultimately, different inversion schemes retained 40 important spatial and spectral features. These 40 features were combined with band features to invert the chlorophyll content of the mangrove canopy.

4) Results and Analysis of Inversion Experiments: The inversion results for each scheme are shown in Fig. 8. Scheme IV demonstrates superior performance, with both models yielding significantly higher estimates (R^2 = 0.718 and 0.759, respectively) compared with other schemes. It shows that the joint inversion of three different source data can provide rich and real spatial spectrum information for the model, enabling the model to better capture patterns and enhance the accuracy of the inversion. Subsequently, the results of the three separate image inversions were compared. Schemes II and III demonstrate significantly higher inversion accuracy compared to Scheme I. The LGBM model achieves superior performance with MR-HSI data, demonstrating a 0.031 increase in R^2 and a 1.88 reduction in RMSE compared with the original image inputs. The R^2 value of the HR-HSI fusion image is 0.042 higher than that of the original image, and the RMSE is 0.44 lower than that of the original image. In Scheme III, the R^2 values of both models were higher than those in Scheme II, indicating that HR-HSI image features have better inversion benefits. This is because after fusing PAN, HR-HSI images have clearer spatial information and can describe the physical characteristics of ground objects more clearly. The inversion results show the powerful benefit of HR-HSI fused images in inversion, and further verify the effectiveness of the GFHMP method and SSICFNet fusion network. Scheme IV achieved higher accuracy improvements in inversion compared with other methods, indicating that the combined reflection method of three different resolution images can combine the differences and advantages between different spatial images to achieve higher accuracy improvements. Fig. 3 of Supple-

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mentary Material displays the thematic maps illustrating the inversion results of mangrove CCC for each proposed scheme.

5) SHAP Interpretive Analysis: Although these inversion methods can effectively predict vegetation CCC, the model prediction process often lacks transparency. To address this challenge, we quantify both global and local feature contributions across all schemes by calculating SHAP values for input variables (see Fig. 9), thereby enabling interpretable analysis of the model's decision-making process. Fig. 9 presents the SHAP summary plots for each scheme, where the horizontal blue bars (left) represent the mean absolute SHAP values, ranked in descending order of global feature importance. The right side shows the local interpretation parameter values for each data point, color-coded according to size. The average SHAP value indicates the overall impact of each feature on the prediction of chlorophyll content in mangrove canopy layers, ranked by importance. The letters L, M, and H in the feature names represent the image features of LR-HSI, MR-HSI, and HR-HSI, respectively. DSI, RSI, NDSI, and NLI denote the distinct vegetation indices for spectral characteristic calculation. The two digits following "B" indicate the utilized spectral bands. For texture features, the numbers after "b" specify the employed bands, while the final letters (M, V, H, C, D, E, S, M, and C2) in feature names correspond to eight texture measures: mean, variance, homogeneity, contrast, dissimilarity, entropy, second moment, and correlation, respectively. Based on the average SHAP value ranking, the most significant feature in Scheme IV is the MR-HSI image feature (M_b7), demonstrating its crucial role in the inversion process. The color mapping reveals that lower values of this feature positively influence the model's predictions. The secondary most influential features are HR-HSI's texture feature (H_b49_C2) and spectral feature (H_RSI_B2_6). Notably, HR-HSI contributes six of the top 12 SHAP-identified features. This predominance stems from the fused image's enhanced spatial clarity and spectral accuracy, which collectively improve inversion precision. In Scheme I, spectral and band features outweigh texture features in importance. This suggests that LR-HSI's low spatial resolution limits its ability to characterize mangrove physiological traits, thereby reducing its effectiveness in CCC inversion. In contrast, Scheme III highlights the elevated importance of texture features, confirming the enhanced spatial information in fused imagery. In Scheme IV, all features exhibit relatively minor negative impacts on model inversion, demonstrating that the joint inversion scheme effectively integrates multiresolution image advantages to enhance inversion accuracy. The results demonstrate that HR-HSI image features in Scheme III exhibit a more pronounced positive effect, significantly enhancing the inversion accuracy of mangrove CCC. In contrast, certain MR-HSI and LR-HSI features in Schemes I and II show inhibitory effects on model predictions.

V. Conclusion

This study proposed the novel GFHMP fusion framework to overcome limitations of existing methods in achieving high-quality, large-scale fusion of satellite hyperspectral data. This framework was compared the performance with current state-of-the-art models in FHM and FHP tasks to determine the optimal model combinations, thereby better achieving information extraction and fusion from satellite hyperspectral, multispectral, and panchromatic data. Under the Wald's protocol, GFHMP significantly reduces training data requirements compared to the conventional one-step fusion of LR-HSI, MR-MSI, and HR-PAN. Furthermore, incorporating MR-MSI provides richer spatial–spectral information while better constraining deep learning-induced image distortions. To maximize GFHMP's performance, we developed SSICFNet, an advanced dual-input fusion network. This network employs multiple convolutional layers to extract both individual and joint features from input images, followed by dense residual connections facilitating multidepth feature extraction. Acknowledging the distinct yet interconnected nature of input modalities, each residual block incorporates a three-way SSICFB. This module computes cross-attention between separate and combined features, enabling better integration of high-frequency and low-frequency information. Experimental results confirmed that SSICFNet surpasses existing state-of-the-art models in extracting spatial and spectral information under Wald's protocol, generating higher quality fused images. SSICFNet achieves optimal performance for both fusion stages within the GFHMP framework. Comprehensive validation, including image fusion, model performance, hyperspectral reconstruction, and mangrove CCC inversion, confirms GFHMP's robustness and SSICFNet's superiority. This article further proposed a joint inversion method that combines the fused image with the original image. Experimental results demonstrated that significant accuracy improvements of 41.60%, 32.92%, and 20.47% compared with inverting individual images. SHAP explainability analysis revealed that the HR-HSI image features contribute more substantially to the inversion model than features from other image types. Furthermore, evaluating feature importance in individual inversions demonstrated the advantage of hyperspectral reconstruction for mangrove CCC retrieval. Although our framework achieves state-of-the-art results on current datasets, its extension to cross-sensor scenarios remains an open challenge for future work, and further exploration of lightweight networks is needed to improve fusion efficiency.

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